

Minimax Dual Certificates for the Common-Action Obstruction

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Abstract

The common-action obstruction certifies that no single instrument serves every branch of an information set; its polyhedral form carries a Farkas multiplier (Abaee 2026a, Section 3.2). This paper states and proves the measure dual of the same obstruction: for control-affine dynamics with a compact convex control set, the common safe-action set is empty precisely when an adversarial probability measure on the active boundary–disturbance bundle drives the expected inward drift strictly negative against every control, with a certificate supported on at most $k+1$ points (k the control dimension; the bound is tight). The polyhedral case recovers the l_1 -normalized Farkas pair exactly; the singleton case recovers the Isaacs drift with the adversarial *distribution* over disturbances load-bearing. A two-action instance exhibits a strict minimax gap: without convexity the equivalence fails and no measure certifies — the audited finite codex systems sit outside this scope by design, certified combinatorially. An obstruction taxonomy (physical, epistemic, institutional) is instantiated on the audited grid, and a calibrated two-stock resource model — one fixed control set throughout — yields a closed-form critical aggregate $Y^* = \frac{27}{5}$: aggregate-yield fibres are viable exactly up to Y^* , the obstruction at $Y = 6$ is certified by a two-point dual measure with margin $\frac{3}{50}$, and the fibre-width formula $\Delta x_1(Y) = Y - 4$ gives the aggregation rule its exact reading. Dynamic-envelope, quantization, and stochastic-bridge extensions are recorded as conjectures; the residual dynamic-observation completeness gap of the source paper remains open. All claims are machine-verified in exact rational arithmetic (8 check families).

Introduction and relation to the programme

The programme’s certificates are emptiness statements: a belief is obstructed when a correspondence of jointly acceptable responses is empty (Abaee 2026a, Theorem 2; Abaee 2026b). For polyhedral safe-action sets the emptiness carries an explicit dual object — the Farkas multiplier vector — so that the obstruction itself becomes a checkable artifact; the sparse-witness bound (at most $m+1$ compatible states) and the singleton Isaacs drift com-

plete the static picture. This paper consolidates the static layer under one roof: the measure dual of the common-action obstruction, of which the normalized Farkas pair, the sparse witness, and the Isaacs reduction are special cases. The formulation was proposed in the programme’s fourth external audit; the present paper verifies its static core exactly, repairs two steps of the proposed argument (the sparsity route and the singleton reading), exhibits the convexity boundary that the proposal’s proof step glossed, adopts its state-versus-epistemic separation axiom, and replaces its worked benchmark — whose evaluation switched control sets between the viable and obstructed fibres — with a single-fixed-control-set model solved in closed form. Nothing in the completed items is modified; the additions are static and computational. The dynamic proposals attached to the same audit are recorded as conjectures in the final section; in particular, the residual dynamic-observation completeness gap (Abaee 2026a, Open Problem 1) is *not* closed here.

The measure dual of the common-action obstruction

Let the safe set be $V = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$ with q_j of class C^1 ; let D be a compact-valued upper semi-continuous disturbance correspondence; let the control set $U \subseteq \mathbb{R}^k$ be compact and *convex*; and let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + G(x)u$, with f_0 and G continuous. For a compact information set $B \subseteq V$ (a fibre or estimation tube) with nonempty active bundle

$$\mathcal{K}(B) = \{(x, j, d) : x \in B, q_j(x) = 0, d \in D(x)\},$$

define the inward drift $\psi(x, j, d; u) = \nabla q_j(x) \cdot [f_0(x, d) + G(x)u]$ and the expected drift $\Phi(u, \mu) = \int_{\mathcal{K}(B)} \psi d\mu$ over Borel probability measures μ on $\mathcal{K}(B)$.

Theorem 1 (minimax dual certificate). Under the standing hypotheses the common safe-action set $\mathcal{R}_V^B(B) = \bigcap_{x \in B} \mathcal{R}_V(x)$ is empty if and only if there exist an adversarial measure $\mu^* \in \mathcal{P}(\mathcal{K}(B))$ and a margin $\varepsilon > 0$ with

$$\max_{u \in U} \Phi(u, \mu^*) \leq -\varepsilon < 0. \quad (\star)$$

Moreover $\max_{u \in U} \min_{\mu \in \mathcal{P}(\mathcal{K}(B))} \Phi(u, \mu) = \min_{\mu} \max_{u \in U} \Phi(u, \mu)$.

Proof. $\mathcal{P}(\mathcal{K}(B))$ is weak-* compact and convex (Banach – Alaoglu), and Φ is continuous, affine in u (control affinity),

and linear in μ ; Sion's minimax theorem applies, and the inner infimum over measures of a linear functional is attained by the Dirac measures, so $\max_u \min_\mu \Phi = \max_u \min_{\mathcal{K}(B)} \psi = \min_\mu \max_u \Phi$. For the equivalence: $u \in \mathcal{R}_V^B(B)$ exactly when $\min_{\mathcal{K}(B)} \psi(x, j, d; u) \geq 0$; the minimum is continuous in u , so nonemptiness of the common safe-action set is equivalent to $\max_{u \in U} \min_{\mathcal{K}(B)} \psi \geq 0$ (the maximum is attained on the compact convex U), and emptiness is equivalent to the common value being strictly negative — which, by the equality, is equivalent to some μ^* with $\max_u \Phi(u, \mu^*) < 0$; the margin is the slack. $\square$

Proposition 1 (atomic support; $k + 1$ witnesses, tight). If $\mathcal{R}_V^B(B) = \emptyset$, the certificate μ^* may be chosen atomic with support of at most $k + 1$ triples, and the bound is tight in dimension k .

Proof. Write $\psi(x, j, d; u) = v(x, j, d) \cdot u + c(x, j, d)$ with $v = G(x)^\top \nabla q_j(x)$ and $c = \nabla q_j(x) \cdot f_0(x, d)$. On the compact set $U \times \mathcal{K}(B)$ the function $-\psi$ is continuous with $\min_u \max_{\mathcal{K}(B)} (-\psi) = 2\eta > 0$ (Theorem 1); a standard finite-cover argument extracts finitely many triples (x_i, j_i, d_i) , $i \leq N$, whose drift rows still satisfy $\min_{u \in U} \max_i (-\psi_i) \geq \eta$. Min-max duality for the finite system (linear programming duality over the simplex) yields $\lambda \geq 0$, $\sum_i \lambda_i = 1$, with $\sum_i \lambda_i \psi_i(u) < 0$ on U ; an extreme such λ is a basic solution of $k + 1$ active constraints and is supported on at most $k + 1$ indices. Tightness: in dimension $k = 2$ the rows $\{u_1 \geq 1\}$, $\{u_2 \geq 1\}$, $\{u_1 + u_2 \leq 1\}$ on the compact square $[-10, 10]^2$ are pairwise intersecting with empty total intersection (checked on the rational grid by the record), so no two witnesses suffice; for $k = 1$ the instance of E1 certifies with exactly two atoms. $\square$

The proposed argument for this proposition mixed Carathéodory's theorem (which yields $k + 2$) with Helly's; the route above — finite reduction, then duality, then the basic-solution support count — is the corrected one, and the record certifies tightness directly.

Proposition 2 (recoveries). (i) *Polyhedral*. If the drift rows are affine with rational data, the measure certificate is the ℓ_1 -normalized Farkas pair of Abaee 2026a, Section 3.2: on the two-floor instance $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{3}{5}\}$ with $U = [0, 1]$, the measure $\lambda = (\frac{1}{2}, \frac{1}{2})$ makes the expected drift *constant* $-\frac{1}{10}$, the normalized multiplier pair of the worked certificate in Section 3.4 there; on the feasible contrast $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{1}{5}\}$ both sides of Theorem 1 equal $+\frac{1}{10}$ and no certificate exists. (ii) *Singleton*. For $B = \{x\}$ the dual measures live on the disturbance set; on the exact convex instance $\psi(u, d_1) = 1 - 2u$, $\psi(u, d_2) = 2u - 1$, $\max_u \min_d \psi = \min_\mu \max_u \Phi = 0$, while the single-worst-disturbance reading $\min_d \max_u \psi = 1$ is *not* the dual: the adversarial distribution over disturbances is load-bearing, and the Isaacs drift is its Dirac-supported inner infimum.

Proposition 3 (convexity boundary). The equivalence of Theorem 1 fails without control convexity. With $U = \{0, 1\}$ and drift rows $(-1, +1)$, $(+1, -1)$, the common safe-action set is empty, yet $\max_u \min \psi = -1$ while $\min_\mu \max_u \Phi = 0$: a strict minimax gap, and no adversarial measure certifies. The audited finite action codices of the catalogue notes are non-convex by construction; there the obstructions are certified combinatorially, and the present theorem is a companion for the continuous, control-affine layer.

The obstruction taxonomy

The standing axiom adopted from the audit: *a dead state is never an information failure*. Three classes of nonviability are distinguished and, on the audited grid, are instantiated side by side (all exact): *physical* — the corner $(1, 1)$ under two

floors is nonviable as a singleton under full information (no instrument keeps both floors); *epistemic* — the audited belief $\{(1, 2), (2, 1)\}$ has every branch viable and the joint set empty (failure by merge); *institutional* — under the unanimity protocol with agency 2 restricted to vote 1, the decentralized kernel collapses from 12 to 0 (failure by protocol; full enumeration). The taxonomy disciplines attribution across the catalogue: mode-blind losses on the same grid are generated entirely at the singleton level (the companion note's Edition 2 record), and no count that mixes the classes is admissible as an epistemic claim.

A calibrated two-stock benchmark

Leaving the audited grid, the continuous layer needs a worked system with one fixed control set throughout. Two renewable stocks ($r_1 = 1$, $r_2 = \frac{4}{5}$, $K_1 = K_2 = 10$, cross-competition $\frac{1}{20}$, shocks $|d_i| \leq \frac{1}{10}$), floors $x_i \geq 2$, and the single control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ (a demand floor; no capacity constraint is imposed). On the aggregate-yield fibre $Y = x_1 + x_2$ within the floors, the active-floor harvest caps are

$$\text{cap}_1(Y) = \frac{3}{2} - \frac{Y-2}{10}, \quad \text{cap}_2(Y) = \frac{59}{50} - \frac{Y-2}{10},$$

so the fibre is viable exactly when $\text{cap}_1(Y) + \text{cap}_2(Y) \geq 2$, i.e.

Theorem 2 (critical aggregate). $Y^* = \frac{27}{5}$: aggregate fibres with $Y \leq \frac{27}{5}$ are viable and those with $Y > \frac{27}{5}$ are obstructed, the value being exact ($\text{cap}_1 + \text{cap}_2 = 2$ at Y^*). At $Y = 5$ the fibre is viable with witness $u = (\frac{6}{5}, \frac{4}{5})$; at $Y = 6$ it is obstructed ($\text{cap sum } \frac{47}{25} < 2$) with dual certificate $\mu^* = (\frac{1}{2}, \frac{1}{2})$ on the two active-floor atoms — two points, within the $k + 1 = 3$ bound — and margin $\varepsilon = \frac{3}{50}$.

The reading is the opposite of the intuition that abundance certifies safety: cross-competition tightens the harvest caps as the aggregate grows, so richer fibres obstruct first — aggregate yield is a failing indicator in both directions. The fibre-width formula makes the aggregation rule exact: the Y -fibre within the floors is $x_1 \in [2, Y - 2]$, of width $\Delta x_1(Y) = Y - 4$, so a linear index cannot separate states closer together than the floor buffer — the composite index is certifying exactly when the width stays inside the buffer, and the certainly-safe reporting of the applications companion governs otherwise.

Conjectures and what is not claimed

Three extensions proposed in the audit are recorded, not claimed. (i) *Dynamic envelope*. A candidate dynamic certificate — an adversarial measure curve $t \mapsto \mu_t^*$ whose expected-margin envelope $Q(t)$ decreases at a rate — is a conjecture: the proposed information-set operator is nonlocal in the state and is not a well-posed Hamilton–Jacobi equation as written; no viscosity well-posedness or comparison argument is supplied; the envelope's crossing of zero is a statement about an expectation, and lifting it to a trajectory violation requires a selection argument through the nonanticipative strategy space that is not provided. In particular, the residual dynamic-observation completeness gap (Abaee 2026a, Open Problem 1) remains open. (ii) *Design formulas*. The dual-margin ratio for safe inspection intervals and the Lipschitz quantization bound are plausible scalings, verified here only instance-wise; they are conjectural as general formulas. (iii) *Stochastic bridge*. The identification of the dual measure with a worst-case prior of the stochastic selector is unproven; the masked-backup mechanics of that lineage must be checked before any bridge is claimed. Scope: continuous-time, control-affine systems with

convex control sets; fibre crossing of the target boundary remains governed by the source paper’s fibre criterion; no completed item is modified; consolidation of the catalogue is an owner-level decision and is not enacted here.

Declarations

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Data availability. No external data were used.
Code availability. The verification script `minimax_dual_certificates_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all eight check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review, under the programme’s verification discipline.